FUNDAMENTAL SI UNITS

DIMENSION	UNIT	SYMBOL
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol

MULTIPLE AND SUBMULTIPLE PREFIXES

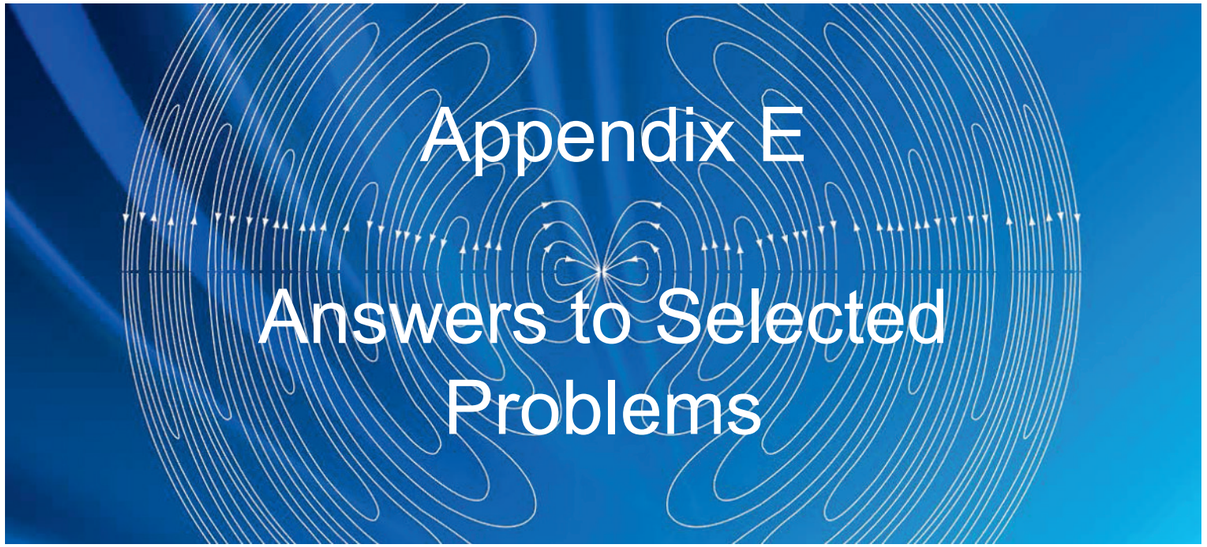
PREFIX	SYMBOL	MAGNITUDE	PREFIX	SYMBOL	MAGNITUDE
exa	E	10^{18}	milli	m	10^{-3}
peta	P	10^{15}	micro	μ	10^{-6}
tera	T	10^{12}	nano	n	10^{-9}
giga	G	10^{9}	pico	p	10^{-12}
mega	M	10^{6}	femto	f	10^{-15}
kilo	k	10^{3}	atto	a	10^{-18}

SI基本单位

量纲	单位	符号
长度	米	m
质量	千克	kg
时间	秒	s
电流	安培	A
温度	开尔文	K
物质的量	摩尔	mol

十进制倍数与分数词头

词头	符号	倍率	词头	符号	倍率
exa	E	10^{18}	毫	m	10^{-3}
拍	P	10^{15}	微	μ	10^{-6}
太	T	10^{12}	纳	n	10^{-9}
吉	G	10^{9}	皮	p	10^{-12}
兆	M	10^{6}	飞	f	10^{-15}
千	k	10^{3}	阿	a	10^{-18}



Appendix E

Answers to Selected Problems

Chapter 1

1.1 10 cm

1.3 $p(x,t) = 32.36\cos(4\pi \times 10^3 t - 12.12\pi x + 36^\circ)$ (N/m^2)

1.6 $u_p = 0.83$ (m/s); $\lambda = 10.47$ m

1.8 **(a)** $y_1(x,t)$ is traveling in positive x direction. $y_2(x,t)$ is traveling in negative x direction.

1.10 $\lambda = 1.4$ m

1.12 $y_2(t)$ lags $y_1(t)$ by 30°.

1.14 $\alpha = 2 \times 10^{-3}$ (Np/m)

1.17 **(c)** $z_1 z_2 = 18e^{j109.4^\circ}$

1.18 **(b)** $z_2 = 1.22(-1+j)$

1.20 **(c)** $|z|^2 = 20$

1.21 **(d)** $t = 0$; $s = 6e^{j30^\circ}$

1.23 $\ln(z) = 1.76 - j1.03$

1.26 $v_c(t) = 15.57\cos(2\pi \times 10^3 t - 81.5^\circ)$ V

1.27 **(d)** $i(t) = 3.61\cos(\omega t + 146.31^\circ)$ A

1.28 **(d)** $\tilde{I} = -2e^{j3\pi/4} = 2e^{-j\pi}e^{j3\pi/4} = 2e^{-j\pi/4}$ A

Chapter 2

2.1 **(a)** $l/\lambda = 1.33 \times 10^{-5}$; transmission line may be ignored.
(c) $l/\lambda = 0.4$; transmission line effects should be included.

2.4 $R' = 1.38$ (Ω/m), $L' = 1.57 \times 10^{-7}$ (H/m), $G' = 0$, $C' = 1.84 \times 10^{-10}$ (F/m)

2.8 $\alpha = 0.109$ Np/m; $\beta = 44.5$ rad/m; $Z_0 = (19.6 + j0.030)$ Ω; $u_p = 1.41 \times 10^8$ m/s

2.10 $w = 1.542$ mm, $\lambda = 0.044$ m

2.14 $R' = 1$ (Ω/m); $L' = 200$ (nH/m); $G' = 400$ (μS/m); $C' = 80$ (pF/m); $\lambda = 2.5$ m

2.16 $R' = 0.6$ Ω/m, $L' = 38.2$ nH/m, $G' = 0.5$ mS/m, $C' = 23.9$ pF/m

2.18 **(a)** $b = 4.2$ mm
(b) $u_p = 2 \times 10^8$ m/s

2.21 $Z_L = (90 - j120)$ Ω

2.23 $Z_0 = 55.9$ Ω

2.27 $Z_{in} = (40 + j20)$ Ω

2.31 **(a)** $\Gamma = 0.62e^{-j29.7^\circ}$

2.32 **(b)** $\Gamma = 0.16e^{-j80.54^\circ}$

2.33 **(a)** $Z_{in_1} = (35.20 - j8.62)$ Ω

2.35 $L = 8.3 \times 10^{-9}$ H

2.37 $l = \lambda/4 + n\lambda/2$

2.39 $Z_{in} = \dfrac{100^2}{33.33} = 300$ Ω

2.41 **(b)** $i_L(t) = 3\cos(6\pi \times 10^8 t - 135^\circ)$ (A)

2.42 **(a)** $Z_{in} = (41.25 - j16.35)$ Ω

2.44 $P_{av}^{i} = 10.0$ mW; $P_{av}^{r} = -1.1$ mW; $P_{av}^{t} = 8.9$ mW

2.45 **(a)** $P_{av} = 0.29$ W

2.47 **(b)** $\Gamma = 0.62e^{-29.7^\circ}$

2.50 $Z_{in} = (66 - j125)$ Ω

2.52 $Z_{01} = 40$ Ω; $Z_{02} = 250$ Ω

2.53 **(b)** $S = 1.64$

2.55 **(a)** $Z_{in} = -j154$ Ω
(b) $0.074\lambda + (n\lambda/2)$, $n = 0, 1, 2, \ldots$

2.57 $y_L = 0.55 + j0.26$

2.61 $Z_L = (41 - j19.5)$ Ω

附录 E

部分习题答案

单位：Np/m 为奈培/米，rad/m 为弧度/米。

习题 2.10 中的 w 表示宽度；公式及分式按原书保留。

校注：由 $\Gamma = (z_L - 1)/(z_L + 1)$，$z_L = 2 - j2$，得 $\Gamma = (7 - j4)/13$；原答案指数漏写 j。

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1.8 (a) $y_1(x,t)$ 沿 +x 方向传播，$y_2(x,t)$ 沿 −x 方向传播。

1.10 $\lambda = 1.4$ m

1.12 $y_2(t)$ 的相位比 $y_1(t)$ 滞后 30°。

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1.28 **(d)** $\tilde{I} = -2e^{j3\pi/4} = 2e^{-j\pi}e^{j3\pi/4} = 2e^{-j\pi/4}$ A

第2章

2.1 (a) $l/\lambda = 1.33 \times 10^{-5}$，可忽略传输线效应；
(c) $l/\lambda = 0.4$，应考虑传输线效应。

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2.57 $y_L = 0.55 + j0.26$

2.61 $Z_L = (41 - j19.5)$ Ω

2.63 $Z_{\text{in}} = (95 - j70)\ \Omega$

2.69 First solution: Stub at $d = 0.199\lambda$ from antenna and stub length $l = 0.125\lambda$. Second solution: $d = 0.375\lambda$ from antenna and stub length $l = 0.375\lambda$.

2.73 $Z_{\text{in}} = 100\ \Omega$

2.78 $V_{\text{g}} = 19.2$ V; $R_{\text{g}} = 30\ \Omega$; $l = 525$ m

2.82 **(a)** $l = 1200$ m

(b) $Z_{\text{L}} = 0$

(c) $R_{\text{g}} = \left(\dfrac{1+\Gamma_{\text{g}}}{1-\Gamma_{\text{g}}}\right) Z_0 = \left(\dfrac{1+0.25}{1-0.25}\right) 50 = 83.3\ \Omega$

(d) $V_{\text{g}} = 32$ V

3.43 $dU/dl = -0.02$

3.47 $\mathbf{E} = \hat{\mathbf{R}} 4R$

3.50 **(a)** $\oint \mathbf{D} \cdot d\mathbf{s} = 150\pi$

(b) $\iiint \nabla \cdot \mathbf{D}\, d\mathcal{V} = 150\pi$

3.58 **(a)** **A** is solenoidal, but not conservative.

(d) **D** is conservative, but not solenoidal.

(h) **H** is conservative, but not solenoidal.

3.59 **(c)** $\nabla^2 \left(\dfrac{3}{x^2+y^2}\right) = \dfrac{12}{(x^2+y^2)^2}$

Chapter 3

3.1 $\hat{\mathbf{a}} = \hat{\mathbf{x}} 0.32 + \hat{\mathbf{z}} 0.95$

3.3 Area $= 36$

3.5 **(a)** $A = \sqrt{14}$; $\hat{\mathbf{a}}_A = (\hat{\mathbf{x}} + \hat{\mathbf{y}} 2 - \hat{\mathbf{z}} 3)/\sqrt{14}$

(e) $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = 20$

(h) $(\mathbf{A} \times \hat{\mathbf{y}}) \cdot \hat{\mathbf{z}} = 1$

3.9 $\hat{\mathbf{a}} = \dfrac{\mathbf{A}}{|\mathbf{A}|} = \dfrac{-\hat{\mathbf{x}} - \hat{\mathbf{y}} y - \hat{\mathbf{z}} 2}{\sqrt{5+y^2}}$

3.11 $\hat{\mathbf{a}} = (\hat{\mathbf{x}} 2 - \hat{\mathbf{z}} 4)/\sqrt{20}$

3.13 $\mathbf{A} = \hat{\mathbf{x}} 0.8 + \hat{\mathbf{y}} 1.6$

3.15 $\hat{\mathbf{c}} = \hat{\mathbf{x}} 0.37 + \hat{\mathbf{y}} 0.56 + \hat{\mathbf{z}} 0.74$

3.17 $\mathbf{G} = \pm\left(-\hat{\mathbf{x}} \frac{8}{3} + \hat{\mathbf{y}} \frac{8}{3} + \hat{\mathbf{z}} \frac{4}{3}\right)$

3.22 **(a)** $P_1 = (2.24,\ 63.4^\circ,\ 0)$ in cylindrical; $P_1 = (2.24,\ 90^\circ,\ 63.4^\circ)$ in spherical

(d) $P_4 = (2.83, 135^\circ, -2)$ in cylindrical; $P_4 = (3.46, 125.3^\circ, 135^\circ)$ in spherical

3.24 **(a)** $P_1 = (0,0,5)$

3.25 **(c)** $A = 12$

3.26 **(a)** $V = 21\pi/2$

3.30 **(a)** $\theta_{AB} = 90^\circ$

(b) $\pm(\hat{\mathbf{r}} 0.487 + \hat{\boldsymbol{\phi}} 0.228 + \hat{\mathbf{z}} 0.843)$

3.32 **(a)** $d = \sqrt{3}$

3.34 **(c)** $\vec{C}(P_3) = \hat{\mathbf{r}} 0.707 + \hat{\mathbf{z}} 4$

(e) $\mathbf{E}(P_5) = -\hat{\mathbf{r}} + \hat{\boldsymbol{\phi}}$

3.35 **(c)** $\mathbf{C}(P_3) = \hat{\mathbf{R}} 0.854 + \hat{\boldsymbol{\theta}} 0.146 - \hat{\boldsymbol{\phi}} 0.707$

3.36 **(e)** $\nabla S = \hat{\mathbf{x}} 8xe^{-z} + \hat{\mathbf{y}} 3y^2 - \hat{\mathbf{z}} 4x^2 e^{-z}$

3.37 **(b)** $\nabla T = \hat{\mathbf{x}} 2x$

(g) $\nabla T = -\hat{\mathbf{x}}\, \frac{2\pi}{6} \sin\left(\frac{\pi x}{3}\right)$

3.38 $T(z) = 10 + (1 - e^{-3z})/3$

3.40 $\left(\frac{dV}{dl}\right)\Big|_{(1,-1,4)} = 2.18$

Chapter 4

4.1 $Q = 2.62$ (mC)

4.3 $Q = 86.65$ (mC)

4.7 $I = 314.2$ A

4.9 **(a)** $\rho_l = -\dfrac{\pi c a^4}{2}$ (C/m)

4.11 $\mathbf{E} = \hat{\mathbf{z}} 51.2$ kV/m

4.13 $q_2 \approx -94.69\ (\mu\text{C})$

4.15 **(a)** $\mathbf{E} = -\hat{\mathbf{x}} 1.6 - \hat{\mathbf{y}} 0.66$ (MV/m)

4.17 $\mathbf{E} = \hat{\mathbf{z}} (\rho_{s0} h/2\varepsilon_0) \left[\sqrt{a^2+h^2} + h^2/\sqrt{a^2+h^2} - 2h\right]$

4.19 $\mathbf{E} = 0$

4.23 **(a)** $\rho_{\text{v}} = y^3 z^3$

(b) $Q = 32$ (C)

(c) $Q = 32$ (C)

4.25 $Q = 4\pi\rho_0 a^3$ (C)

4.27 $\mathbf{D} = \hat{\mathbf{r}} \dfrac{\rho_{\text{v}0}(r^2-1)}{2r}, \quad 1 \le r \le 3$ m

$\mathbf{D} = \hat{\mathbf{r}} D_r = \hat{\mathbf{r}} \dfrac{4\rho_{\text{v}0}}{r}, \quad r \ge 3$ m

4.30 $R_1 = \dfrac{a}{2},\ R_3 = \dfrac{a\sqrt{5}}{2},\ V = \dfrac{0.55Q}{\pi\varepsilon_0 a}$

4.32 **(b)** $\mathbf{E} = \hat{\mathbf{z}} (\rho_l a/2\varepsilon_0)[z/(a^2+z^2)^{3/2}]$ (V/m)

4.34 $V(b) = \dfrac{\rho_l}{4\pi\varepsilon} \ln\left[\dfrac{l + \sqrt{l^2+4b^2}}{-l + \sqrt{l^2+4b^2}}\right]$ (V)

4.37 $V = \dfrac{\rho_l}{2\pi\varepsilon_0}\left[\ln\left(\dfrac{a}{\sqrt{(x-a)^2+y^2}}\right) - \ln\left(\dfrac{a}{\sqrt{(x+a)^2+y^2}}\right)\right]$

4.39 $V_{AB} = -117.09$ V

4.41 **(c)** $\mathbf{u}_{\text{e}} = -8.125\mathbf{E}/|\mathbf{E}|$ (m/s); $\mathbf{u}_{\text{h}} = 3.125\mathbf{E}/|\mathbf{E}|$ (m/s)

4.45 $R = 4.2\ (\text{m}\Omega)$

4.48 $\theta = 61^\circ$

2.63 $Z_{\text{in}} = (95 - j70)\ \Omega$

2.69 第一组解：短路支节距天线 d = 0.199λ，支节长度 l = 0.125λ。第二组解：d = 0.375λ，l = 0.375λ。

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(b) $\iiint \nabla \cdot \mathbf{D}\, d\mathcal{V} = 150\pi$

3.58 (a) A 为无散场，但非保守场。
(d) D 为保守场，但非无散场。
(h) H 为保守场，但非无散场。

3.59 **(c)** $\nabla^2 \left(\dfrac{3}{x^2+y^2}\right) = \dfrac{12}{(x^2+y^2)^2}$

第3章

3.1 $\hat{\mathbf{a}} = \hat{\mathbf{x}} 0.32 + \hat{\mathbf{z}} 0.95$

3.3 面积 = 36

3.5 **(a)** $A = \sqrt{14}$; $\hat{\mathbf{a}}_A = (\hat{\mathbf{x}} + \hat{\mathbf{y}} 2 - \hat{\mathbf{z}} 3)/\sqrt{14}$

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(d) P_4：圆柱坐标 (2.83, 135°, −2)；球坐标 (3.46, 125.3°, 135°)。

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4.45 $R = 4.2\ (\text{m}\Omega)$

4.48 $\theta = 61^\circ$

4.50 $Q = \dfrac{3\pi\varepsilon_0}{2}$ (C)

4.53 **(a)** $|E|$ is maximum at $r = a$.

4.55 $W_e = 4.62 \times 10^{-9}$ (J)

4.57 **(a)** $C = 3.1$ pF

4.60 **(b)** $C = 6.07$ pF

4.63 $C' = \dfrac{\pi\varepsilon_0}{\ln[(2d/a) - 1]}$ (C/m)

Chapter 5

5.1 $\mathbf{a} = -\hat{\mathbf{y}}4.22 \times 10^{18}$ (m/s^2)

5.4 $\vec{T} = -\hat{\mathbf{z}}1.66$ (N·m); clockwise

5.5 **(a)** $\mathbf{F} = 0$

5.7 $\mathbf{B} = -\hat{\mathbf{z}}0.6$ (mT)

5.9 $\mathbf{H} = \hat{\mathbf{z}}\dfrac{I\theta\,(b-a)}{4\pi ab}$

5.11 $I_2 = \dfrac{2aI_1}{2\pi Nd} = \dfrac{1 \times 25}{\pi \times 20 \times 2} = 0.2$ A

5.13 $I = 200$ A

5.16 $\vec{F} = -\hat{\mathbf{x}}0.4$ (mN)

5.18 **(a)** $\mathbf{H}(0,0,h) = -\hat{\mathbf{x}}\,\dfrac{I}{\pi w}\tan^{-1}\left(\dfrac{w}{2h}\right)$ (A/m)

5.20 $\mathbf{F} = \hat{\mathbf{y}}4 \times 10^{-5}$ N

5.24 $\mathbf{J} = \hat{\mathbf{z}}36e^{-3r}$ A/m^2

5.26 **(a)** $\vec{A} = \hat{\mathbf{z}}\,\dfrac{\mu_0 I}{4\pi}\ln\left(\dfrac{\ell + \sqrt{\ell^2 + 4r^2}}{-\ell + \sqrt{\ell^2 + 4r^2}}\right)$

5.27 **(a)** $\mathbf{B} = \hat{\mathbf{z}}5\pi\sin\pi y - \hat{\mathbf{y}}\pi\cos\pi x$ (T)

5.29 **(a)** $\mathbf{A} = \hat{\mathbf{z}}\mu_0 IL/(4/piR)$

(b) $\mathbf{H} = (IL/4\pi)[(-\hat{\mathbf{x}}y + \hat{\mathbf{y}}x)/(x^2 + y^2 + z^2)^{3/2}]$

5.31 $n_e = 1.5$ electrons/atom

5.33 $\mathbf{H}_2 = \hat{\mathbf{z}}3$

5.35 $\vec{B}_2 = \hat{\mathbf{x}}20{,}000 - \hat{\mathbf{y}}30{,}000 + \hat{\mathbf{z}}8$

5.37 $L' = (\mu/\pi)\ln[(d-a)/a]$ (H)

5.40 $\Phi = 1.66 \times 10^{-6}$ (Wb)

Chapter 6

6.1 At $t = 0$, current in top loop is momentarily clockwise. At $t = t_1$, current in top loop is momentarily counterclockwise.

6.3 **(a)** $V_{emf} = 375e^{-3t}$ (V)

6.5 $B_0 = 0.8$ (nA/m)

6.7 $I_{ind} = 37.7\sin(200\pi t)$ mA

6.10 $V_{12} = -236$ (μV)

6.12 $I = 0.1$ (A)

6.15 $I = 0.82\cos(120\pi t)$ (μA)

6.17 $f = 5$ MHz

6.18 **(b)** 888

6.20 $\rho_v = (8y/\omega)\sin\omega t + C_0$, where C_0 is a constant of integration.

6.24 $k = (4\pi/30)$ rad/m;
$\mathbf{E} = -\hat{\mathbf{z}}941\cos(2\pi \times 10^7 t + 4\pi y/30)$ (V/m)

6.26 $\mathbf{H}(R,\theta;t) = \hat{\boldsymbol{\phi}}(53/R)\sin\theta\cos(6\pi \times 10^8 t - 2\pi R)$ (μA/m)

6.28 **(a)** $k = 20$ (rad/m)

Chapter 7

7.1 **(a)** Positive y direction

(c) $\lambda = 12.6$ m

7.3 **(a)** $\lambda = 31.42$ m

7.5 $\varepsilon_r = 9$

7.7 **(a)** $\lambda = 20$ m

7.9 $\mathbf{E} = \hat{\mathbf{x}}\sqrt{2}\cos(\omega t + kz) - \hat{\mathbf{y}}\sqrt{2}\sin(\omega t + kz)$ (V/m)

7.12 At $x = 0$ and $t = 0$, $\mathbf{E} = -\hat{\mathbf{z}}1.885$ (V/m)
At $x = 0$ and $t = 5$ ns, $\omega t = 0.13$ rad and
$\mathbf{E} = -1.885(\hat{\mathbf{y}}0.13 + \hat{\mathbf{z}}0.99)$ (V/m)

7.14 **(a)** $\gamma = 73.5°$ and $\chi = -8.73°$

(b) Right-hand elliptically polarized

7.17 **(a)** Low-loss dielectric. $\alpha = 8.42 \times 10^{-11}$ Np/m, $\beta = 468.3$ rad/m, $\lambda = 1.34$ cm, $u_p = 1.34 \times 10^8$ m/s, $\eta_c \approx 168.5\ \Omega$

7.19 **H** lags **E** by 31.72°

7.21 $z = 287.82$ m

7.23 $u_p = 9.42 \times 10^4$ (m/s)

7.25 $\mathbf{H} = -\hat{\mathbf{y}}0.16e^{-30x}\cos(2\pi \times 10^9 t - 40x - 36.85°)$ (A/m)

7.29 $(R_{ac}/R_{dc}) = 143.55$

7.33 $\mathbf{S}_{av} = \hat{\mathbf{y}}0.48$ (W/m^2)

7.35 **(c)** $z = 23.03$ m

7.37 $u_p = 1 \times 10^8$ (m/s)

7.39 **(b)** $P_{av} = 0$

7.41 **(a)** $(w_e)_{av} = \dfrac{\varepsilon E_0^2}{4}$

4.50 $Q = \dfrac{3\pi\varepsilon_0}{2}$ (C)

4.53 (a) |E| 在 r = a 处达到最大值。

4.55 $W_e = 4.62 \times 10^{-9}$ (J)

4.57 **(a)** $C = 3.1$ pF

4.60 **(b)** $C = 6.07$ pF

4.63 $C' = \dfrac{\pi\varepsilon_0}{\ln[(2d/a) - 1]}$ (C/m)

第5章

5.1 $\mathbf{a} = -\hat{\mathbf{y}}4.22 \times 10^{18}$ (m/s^2)

5.4 $\vec{T} = -\hat{\mathbf{z}}1.66$ (N·m); （顺时针）

5.5 **(a)** $\mathbf{F} = 0$

5.7 $\mathbf{B} = -\hat{\mathbf{z}}0.6$ (mT)

5.9 $\mathbf{H} = \hat{\mathbf{z}}\dfrac{I\theta\,(b-a)}{4\pi ab}$

5.11 $I_2 = \dfrac{2aI_1}{2\pi Nd} = \dfrac{1 \times 25}{\pi \times 20 \times 2} = 0.2$ A

5.13 $I = 200$ A

5.16 $\vec{F} = -\hat{\mathbf{x}}0.4$ (mN)

5.18 **(a)** $\mathbf{H}(0,0,h) = -\hat{\mathbf{x}}\,\dfrac{I}{\pi w}\tan^{-1}\left(\dfrac{w}{2h}\right)$ (A/m)

5.20 $\mathbf{F} = \hat{\mathbf{y}}4 \times 10^{-5}$ N

5.24 $\mathbf{J} = \hat{\mathbf{z}}36e^{-3r}$ A/m^2

5.26 **(a)** $\vec{A} = \hat{\mathbf{z}}\,\dfrac{\mu_0 I}{4\pi}\ln\left(\dfrac{\ell + \sqrt{\ell^2 + 4r^2}}{-\ell + \sqrt{\ell^2 + 4r^2}}\right)$

5.27 **(a)** $\mathbf{B} = \hat{\mathbf{z}}5\pi\sin\pi y - \hat{\mathbf{y}}\pi\cos\pi x$ (T)

5.29 (a) $A_z = \mu_0 IL/(4\pi R)$，其余分量为零。

(b) $\mathbf{H} = (IL/4\pi)[(-\hat{\mathbf{x}}y + \hat{\mathbf{y}}x)/(x^2 + y^2 + z^2)^{3/2}]$

5.31 n_e = 1.5 个电子/原子

5.33 $\mathbf{H}_2 = \hat{\mathbf{z}}3$

5.35 $\vec{B}_2 = \hat{\mathbf{x}}20{,}000 - \hat{\mathbf{y}}30{,}000 + \hat{\mathbf{z}}8$

5.37 $L' = (\mu/\pi)\ln[(d-a)/a]$ (H)

5.40 $\Phi = 1.66 \times 10^{-6}$ (Wb)

第6章

6.1 t = 0 时上方回路电流沿顺时针方向；t = t_1 时沿逆时针方向。

6.3 **(a)** $V_{emf} = 375e^{-3t}$ (V)

6.5 $B_0 = 0.8$ (nA/m)

6.7 $I_{ind} = 37.7\sin(200\pi t)$ mA

6.10 $V_{12} = -236$ (μV)

6.12 $I = 0.1$ (A)

6.15 $I = 0.82\cos(120\pi t)$ (μA)

6.17 $f = 5$ MHz

6.18 **(b)** 888

6.20 $\rho_v = (8y/\omega)\sin\omega t + C_0$, C_0 为积分常数。

6.24 $k = (4\pi/30)$ rad/m;
$\mathbf{E} = -\hat{\mathbf{z}}941\cos(2\pi \times 10^7 t + 4\pi y/30)$ (V/m)

6.26 $\mathbf{H}(R,\theta;t) = \hat{\boldsymbol{\phi}}(53/R)\sin\theta\cos(6\pi \times 10^8 t - 2\pi R)$ (μA/m)

6.28 **(a)** $k = 20$ (rad/m)

第7章

7.1 (a) 沿 +y 方向

(c) $\lambda = 12.6$ m

7.3 **(a)** $\lambda = 31.42$ m

7.5 $\varepsilon_r = 9$

7.7 **(a)** $\lambda = 20$ m

7.9 $\mathbf{E} = \hat{\mathbf{x}}\sqrt{2}\cos(\omega t + kz) - \hat{\mathbf{y}}\sqrt{2}\sin(\omega t + kz)$ (V/m)

7.12 当 $x = 0$ 且 $t = 0$, $\mathbf{E} = -\hat{\mathbf{z}}1.885$ (V/m)
当 $x = 0$ 且 $t = 5$ ns, $\omega t = 0.13$ rad 且
$\mathbf{E} = -1.885(\hat{\mathbf{y}}0.13 + \hat{\mathbf{z}}0.99)$ (V/m)

7.14 **(a)** $\gamma = 73.5°$ and $\chi = -8.73°$

(b) 右旋椭圆极化

7.17 **(a)** 低损耗电介质； $\alpha = 8.42 \times 10^{-11}$ Np/m, $\beta = 468.3$ rad/m, $\lambda = 1.34$ cm, $u_p = 1.34 \times 10^8$ m/s, $\eta_c \approx 168.5\ \Omega$

7.19 H 的相位比 E 滞后 31.72°。

7.21 $z = 287.82$ m

7.23 $u_p = 9.42 \times 10^4$ (m/s)

7.25 $\mathbf{H} = -\hat{\mathbf{y}}0.16e^{-30x}\cos(2\pi \times 10^9 t - 40x - 36.85°)$ (A/m)

7.29 $(R_{ac}/R_{dc}) = 143.55$

7.33 $\mathbf{S}_{av} = \hat{\mathbf{y}}0.48$ (W/m^2)

7.35 **(c)** $z = 23.03$ m

7.37 $u_p = 1 \times 10^8$ (m/s)

7.39 **(b)** $P_{av} = 0$

7.41 **(a)** $(w_e)_{av} = \dfrac{\varepsilon E_0^2}{4}$

校注：5.29 按短电流元的磁矢势公式修正原书分母中的“/pi”误印。

Chapter 8

8.1 (a) $\Gamma = -0.67;\ \tau = 0.33$
(b) $S = 5$
(c) $S_{av}^{i} = 0.52\ (\text{W/m}^2);\ S_{av}^{r} = 0.24\ (\text{W/m}^2);$
$S_{av}^{t} = 0.28\ (\text{W/m}^2)$

8.3 (b) $\mathbf{S}_{av}^{i} = \hat{\mathbf{y}}251.34,\ \mathbf{S}_{av}^{r} = \hat{\mathbf{y}}10.05,$
$\mathbf{S}_{av}^{t} = \hat{\mathbf{y}}241.29 \quad (\text{W/m}^2)$

8.6 (a) $\Gamma = -0.71$

8.7 $|\widetilde{\mathbf{E}}_1|_{\max} = 85.5\ (\text{V/m});\ l_{\max} = 1.5\ \text{m}$

8.9 $\varepsilon_{r_2} = \sqrt{\varepsilon_{r_1}\varepsilon_{r_3}};\ d = c/[4f(\varepsilon_{r_1}\varepsilon_{r_3})^{1/4}]$

8.11 $Z_{in}(-d) = 0.43\eta_0\underline{/-51.7^\circ}$
$|\Gamma|^2 = 0.24$

8.13 $f = 75\ \text{MHz}$

8.15 $P' = (3.3\times10^{-3})^2\frac{10^2}{2}\times1.14[1-e^{-2\times44.43\times2\times10^{-3}}]$
$= 1.01\times10^{-4} \quad (\text{W/m}^2)$

8.17 $\theta_{\min} = 20.4^\circ$

8.19 $\dfrac{S^t}{S^i} = 0.85$

8.22 $d = 15\ \text{cm}$

8.24 $d = 68.42\ \text{cm}$

8.26 $f_p = 59.88\ (\text{Mb/s})$

8.29 (b) $\theta_i = 36.87^\circ$

8.31 (a) $\theta_i = 33.7^\circ$

8.33 $\theta_t = 18.44^\circ$

8.39 (a) $R = 6.4\times10^{-3};\ T = 0.9936$
(b) $P^i = 85\ \text{mW};\ P^r = 0.55\ \text{mW};\ P^t = 84.45\ \text{mW}$

8.42 (a) 9.4%

8.47 $a = 3.33\ \text{cm};\ b = 2\ \text{cm}$

8.49 Any one of the first four modes.

8.51 570 Ω (empty); 290 Ω (filled)

8.53 $\theta'_{20} = 57.7^\circ$

8.55 (a) $Q = 8367$

Chapter 9

9.1 $S_{\max} = 7.6\ (\mu\text{W/m}^2)$

9.4 (a) Direction of maximum radiation is a circular cone 120° wide, centered around the $+z$ axis.
(b) $D = 4 = 6\ \text{dB}$
(c) $\Omega_p = \pi\ (\text{sr}) = 3.14\ (\text{sr})$
(d) $\beta = 120^\circ$

9.6 (b) $G = -3.5\ \text{dB}$

9.9 $S_{\max} = 4\times10^{-5}\ (\text{W/m}^2)$

9.11 $D = 36.61\ \text{dB}$

9.14 $S = 1.46$

9.16 (a) $\widetilde{\mathbf{E}}(R,\theta,\phi) = \hat{\boldsymbol{\theta}}\widetilde{E}_\theta = \hat{\boldsymbol{\theta}}j\,\dfrac{I_0 lk\eta_0}{8\pi}\left(\dfrac{e^{-jkR}}{R}\right)\sin\theta\ (\text{V/m})$

9.17 (a) $\theta_{\max_1} = 42.6^\circ,\ \theta_{\max_2} = 137.4^\circ$

9.20 (a) $\theta_{\max_1} = 90^\circ,\ \theta_{\max_2} = 270^\circ$
(b) $S_{\max} = \dfrac{60I_0^2}{\pi R^2}$
(c) $F(\theta) = \dfrac{1}{4}\left[\dfrac{\cos(\pi\cos\theta)+1}{\sin\theta}\right]^2$

9.26 $P_t = 259\ (\text{mW})$

9.28 $P_t = 75\ (\mu\text{W})$

9.30 (a) $P_{rec} = 3.6\times10^{-6}\ \text{W}$

9.33 $\beta_{null} = 5.73^\circ$

9.35 $D = 45.6\ \text{dB}$

9.38 (a) $\beta_e = 1.8^\circ;\ \beta_a = 0.18^\circ$
(b) $\Delta y = \beta_a R = 0.96\ \text{m}$

9.42 (a) $F_a(\theta) = 4\cos^2\left[\frac{\pi}{8}(4\cos\theta+1)\right]$

9.44 $d/\lambda = 1.414$

9.49 $F_a(\theta) = [6+8\cos(\pi\cos\theta)+2\cos(2\pi\cos\theta)]^2$

9.51 $\delta = -2.72\ (\text{rad}) = -155.9^\circ$

Chapter 10

10.1 $T = 107.26\ \text{minutes}$

10.3 $133.3 \approx 133\ \text{channels}$

10.5 $(f_p)_{\max} = 300\ \text{kHz}$

10.7 $R_{\max} = 4.84\ \text{km}$

第8章

8.1 (a) $\Gamma = -0.67;\ \tau = 0.33$
(b) $S = 5$
(c) $S_{av}^{i} = 0.52\ (\text{W/m}^2);\ S_{av}^{r} = 0.24\ (\text{W/m}^2);$
$S_{av}^{t} = 0.28\ (\text{W/m}^2)$

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8.39 (a) $R = 6.4\times10^{-3};\ T = 0.9936$
(b) $P^i = 85\ \text{mW};\ P^r = 0.55\ \text{mW};\ P^t = 84.45\ \text{mW}$

8.42 (a) 9.4%

8.47 $a = 3.33\ \text{cm};\ b = 2\ \text{cm}$

8.49 前四个模式中的任意一个。

8.51 空腔：570 Ω；填充介质后：290 Ω。

8.53 $\theta'_{20} = 57.7^\circ$

8.55 (a) $Q = 8367$

第9章

9.1 $S_{\max} = 7.6\ (\mu\text{W/m}^2)$

9.4 (a) 最大辐射方向构成绕 +z 轴的圆锥面，圆锥全顶角为 120°。
(b) $D = 4 = 6\ \text{dB}$
(c) $\Omega_p = \pi\ (\text{sr}) = 3.14\ (\text{sr})$
(d) $\beta = 120^\circ$

9.6 (b) $G = -3.5\ \text{dB}$

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9.44 $d/\lambda = 1.414$

9.49 $F_a(\theta) = [6+8\cos(\pi\cos\theta)+2\cos(2\pi\cos\theta)]^2$

9.51 $\delta = -2.72\ (\text{rad}) = -155.9^\circ$

第10章

10.1 T = 107.26 min

10.3 133.3，即约 133 个信道。

10.5 $(f_p)_{\max} = 300\ \text{kHz}$

10.7 $R_{\max} = 4.84\ \text{km}$